

JavaFX: A Rich Internet Application (RIA) Development Platform

JavaFX, with its rich set of APIs and Oracle's committed support, is a natural choice for high performance, data-centric rich Internet applications. A vibrant JavaFX community continues to develop and support JavaFX porting on mobile platforms.



Write once, run everywhere (WORA) was the Java punch line for decades, and it helped companies save millions of dollars which otherwise would have been spent on porting the same application on different platforms. This capability was part of the reason behind Java's unprecedented success. Java toolkits for user interface (UI) development also follow the same pattern. The AWT library introduced in 1995 for UI development was based on the same philosophy. This library was dependent on native calls for drawing objects. Native dependency in a multi-threading environment caused lock contentions, leading to deadlocks and heavy utilisation of system resources. The look and feel of such applications depended on the native platform and was not uniform across platforms.

Swing was the next UI toolkit to address these issues. Swing became very popular and was a platform of choice for desktop application development. This continued till the popularity of Web applications caught on, changing the way a UI was used and designed. A large number of desktop applications in manufacturing, automotive, banking, software development IDEs and the healthcare domain continue to largely use Swing based applications. However, Swing uses the event dispatch thread for both event dispatch and

rendering, thus limiting the hardware acceleration benefits on modern multi-core systems and also lacks efficient 3D libraries, animation, charts and media.

As we progressed in time, use of browser-based Web applications increased tremendously and other computing devices such as tablets, smartphones, set-top boxes, etc., became more powerful in terms of processing power and memory. The changed scenario demanded a common development platform which could integrate rich Internet features into the desktop environment and support a cross-platform, cross-device programming environment. JavaFX was the answer to this requirement.

JavaFX 1.0 was launched by Sun Microsystems (acquired by Oracle in 2010) in 2008 as the next generation UI toolkit and aimed to become the successor of Swing. It was designed to leverage new age GPUs with hardware acceleration capabilities and added support for 3D, animation, media, WebView and CSS style. JavaFX1.0 was a script based language for the Java platform that could interoperate with it. JavaFX 2.0 was completely based on the Java API allowing any Java developer to use it with the IDE. JavaFX by Oracle is available for desktop application development and on mobile devices by community support.